Forecasting Generative AI Work Capabilities with a LLM Dynamic Benchmark

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Discussion by Simon Bunel (Banque de France and ENS)

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Any opinions and conclusions expressed in this discussion do not necessarily represent the views of the Banque de France.

This paper: What it does

- Provides empirical insights into LLM capabilities for performing workplace-related human tasks
- Designs practical exams generated and evaluated entirely by an LLM, then taken by another LLM
- Focuses on core tasks in three major occupation groups:
  - Business & Financial Operations
  - Computer & Mathematical Occupations
  - Management Occupations
- Builds a subset of 143 tasks where credible practical exams can be designed (text-only interaction, no external tools)
- Generates an exam for each task and evaluates multiple LLMs on their performance

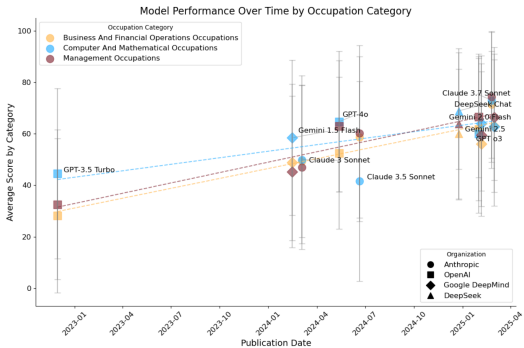
This paper: Why is it important?

- On the future of work:
  - Estimates of employment *exposure to AI* in advanced economies rely on task- or ability-level exposure
    - ⋆ Typically based on expert or LLM-generated judgments, which are difficult to validate empirically
    - ⋆ Resulting estimates vary widely: 18.5% (Gmyrek et al., 2023); 19.9% (Eloundou et al., 2023); up to 60% (Pizzinelli et al., 2023), among others.
- On the future of productivity growth:
  - Acemoglu (2025) proposes the following framework for estimating TFP gains over 10 years:

$$\begin{aligned}\text{TFP gains} = {} & \text{GDP share of AI-exposed tasks} \\ & \times \text{Share of exposed tasks where AI adoption is economically profitable} \\ & \times \text{Average cost savings from AI} \\ & \times \text{AI exposure-adjusted labor share}\end{aligned}$$

  - Given the existing literature, estimates can vary by a factor of three when considering only this first component
  - This paper provides empirically grounded estimates to help assess the potential productivity gains from AI

This paper: What it finds



- For Business & Financial Operations, Management, and Computer & Mathematical occupations, only 5–10% of tasks can currently be tested using text-only LLMs
- Among this subset, state-of-the-art models achieve average performance scores between 63% and 73%
- Progress over time: older models like GPT-3.5 Turbo (from two years ago) scored 32% on average
- Suggests a roughly linear improvement trajectory

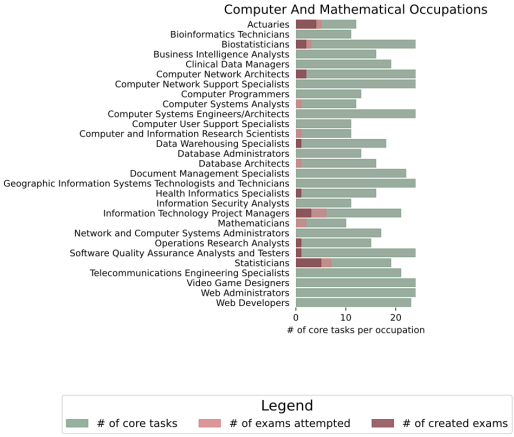
General Comment

- Excellent paper on a highly relevant topic!
- Key contributions:
  - Offers a descriptive analysis of LLM capability evolution over the past two years
  - Introduces a structured framework to document the evolution of LLM capabilities and their interaction with human tasks in the coming years
- To the best of my knowledge, this is the first paper to adopt a systematic task-testing approach
- Strong policy relevance: Implications for understanding the impact of AI on labor markets and productivity

Comment #1: Representativeness

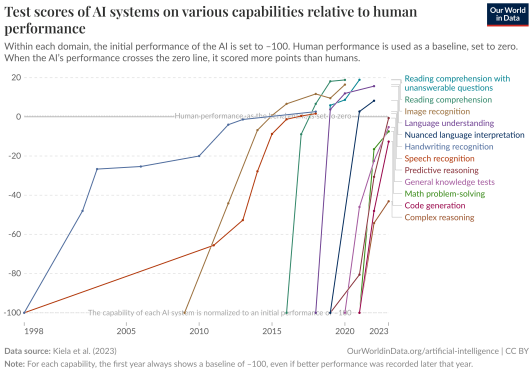
- The paper focuses on core tasks within three major occupational groups: Business & Financial Operations, Computer & Mathematical Occupations, and Management Occupations
  - Could you elaborate on the motivation for selecting these specific groups?
  - It would be helpful to better understand how representative these occupations are within the labor market (employment share, wages, etc.)

Comment #1: Representativeness



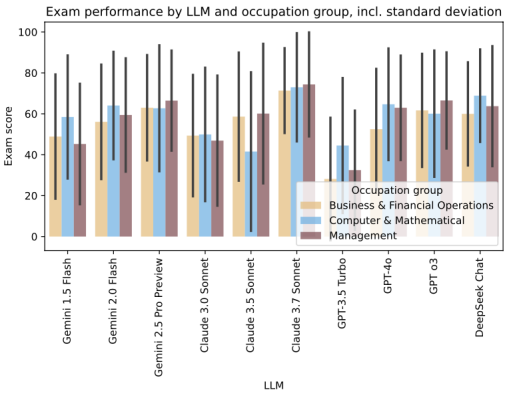
- Within the selected occupations, only 5% to 10% of tasks are currently tested
- This excludes multimodal capabilities (e.g., image or audio processing) and tool use (e.g., coding, manipulating spreadsheets)
- Could the focus on text-based interactions introduce a bias in assessing current LLM capabilities?
- For instance, Peng et al. (2023) report productivity gains of $+55.8\%$ for programmers

Comment #1: Representativeness



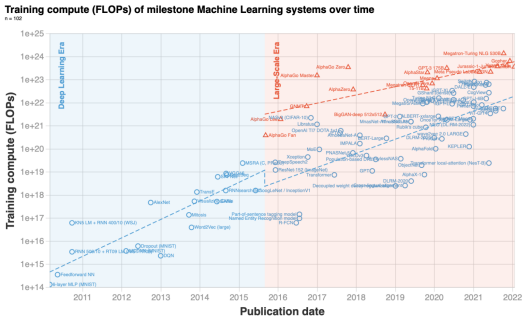
- It could also affect the assessment of capability evolution, as LLMs may progress at different speeds depending on the type of task

Comment #2: Heterogeneity of the Effects & Extension



- Sizable standard deviation within occupation groups suggests heterogeneity in exam scores at the task level
- Would be interesting to explore whether model improvements are consistent over time - do models continue to perform best on the same tasks as they evolve?
- This framework seems well suited to investigate that
- One possible approach: rank-rank regression to examine the stability of task-level performance rankings over time

Comment #3: What about the future?

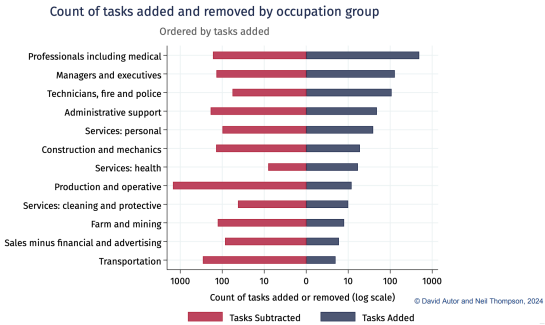


Source : Sevilla et al. (2022)

- Can we extrapolate the future evolution of LLMs based on recent performance trends?
- The paper is careful not to overclaim
- AI progress has sometimes occurred through sudden breakthroughs rather than continuous improvement

  $\longrightarrow$ Approach complementary to forward-looking methods (e.g., Felten et al., 2021; Eloundou et al., 2024)

Comment #3: What about the future?



Source : Autor and Thomson (2024)

- This framework does not capture the potential (probable?) emergence of new tasks in the future
- Autor and Thomson (2024): between 1977 and 2018, many occupations experienced not only task displacement but also substantial task creation